EN211:COMMUNITY ENGAGEMENT AND ENTREPRENEURSHIP PROGRAM

Solar-Powered Smart Bus Stops with Wi-Fi & Safety Features

1. Problem Statement

Urban areas in India face growing challenges with unstructured public transportation infrastructure. Most bus stops lack essential services such as lighting, real-time schedule updates, security, and shelter, making them unsafe and inaccessible. This is especially critical for women, children, and the elderly.

The absence of digital connectivity and security at bus stops also limits the effectiveness of smart city initiatives and public transport systems. Additionally, conventional bus shelters rely on grid electricity, adding to urban carbon footprints.

This project supports SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 11 (Sustainable Cities and Communities) through the implementation of a solar-powered smart bus stop with enhanced safety and digital connectivity.

2.Novelty Statement

This project is presenting a first-of-its-kind, integrated solar-powered intelligent bus stop system that integrates all key functionalities for contemporary urban commutes within a single, sustainable framework. In contrast to current models that merely employ solitary functions—like solar panels, CCTV, or digital screens—this model supports a complete set: solar energy independence, CCTV monitoring, public Wi-Fi, USB charging stations, real-time digital screens, panic button, EV charging capability, and 10 accessible seating under a weather-sheltered design.

3. Objectives

- To design and build a solar-powered smart bus stop with seating for 10 people.
- To integrate features like Wi-Fi, CCTV, panic button, LED lighting, and USB charging.
- To ensure complete energy self-sufficiency using solar power and battery storage.
- To promote safety, digital access, and environmental sustainability.

4. Existing Bus Stop Infrastructure: A Critical Review

The majority of India's and other developing countries traditional bus stops are simple structures or just marked roadside areas. These stops are typically constructed without

regard to user comfort, safety, accessibility, or sustainability. The major problems with these traditional bus stops are:

- No lighting: Makes stops unsafe at night, particularly for women and the elderly.
- No seating or shelter: Passengers are exposed to sun, rain, or dust while waiting.
- Lack of real-time information: No electronic display or public signage for timings.
- No safety features: No CCTV, panic buttons, or security personnel presence.
- Unreliable power dependence: Complete reliance on grid electricity, which is usually not available.
- Accessibility constraints: No ramps or arrangements for differently-abled commuters.

This absence of user-focused design diminishes public transport attractiveness and effectiveness. A new, intelligent, solar-based solution is required to address increasing urban needs

Figure : bus stops in some rural areas

Certain cities have also made efforts towards upgrading their bus stops. Bengaluru, Pune, and some areas of Delhi have experimented with or implemented newer bus stops that include solar lighting, CCTV, or simple digital display boards. These are scattered, non-standardized, and do not come with a unified smart solution. The majority of the nation's bus stops remain old, necessitating a smart model that is scalable and integrated.

Figure : Bus stops that have only a few features

Top Cities for Implementation

Bengaluru (Tech Hub & Public Transport Demand) Why?

- High population density & heavy bus usage (BMTC handles ~35 lakh passengers daily).
- Existing smart bus stops in select areas – ideal for full-scale expansion. Strong IT infrastructure & public adoption of QR payments, UPI & digital systems.
- High solar energy potential for powering smart bus stops.

Delhi (Capital & Smart City Initiatives) Why?

- Largest public transport network with DTC & cluster buses (~45 lakh daily ridership).
- Delhi Metro has integrated smart ticketing & bus tracking – easy expansion to bus stops.
- High investment in urban infrastructure & smart city projects. Severe air pollution issues – solar bus stops can promote sustainability.

Mumbai (Heavy Commuter Traffic & Financial Capital) Why?

- BEST buses serve ~30 lakh passengers daily, requiring smart bus stops for efficiency. Existing digital ticketing & QR-based payments.
- High demand for real-time tracking & crowd monitoring due to congestion.
- Solar energy can reduce reliance on grid power, helping energy efficiency.

Pune (Smart City with Strong EV & Solar Push) Why?

- Pune Smart City Mission has already introduced smart bus stops – ideal for scaling up.
- Growing adoption of EVs & sustainable infrastructure.
- High WiFi adoption & digital literacy, making public WiFi at bus stops more impactful.

Hyderabad (Growing Metro & Bus Integration) Why?

- TSRTC buses serve ~35 lakh passengers daily, requiring smart facilities.
- Metro & bus connectivity makes smart ticketing & digital information useful.
- Government initiatives promoting solar & smart infrastructure (Expanding Public Transport & Renewable Energy Adoption)

Chennai Why?

- MTC buses have ~25 lakh daily commuters – high demand for smart services.
- State government pushing for solar & sustainable infrastructure.
- Coastal heat makes solar-powered cooling & shelters highly beneficial.

To implement smart solar-powered bus stops, you can leverage the following government schemes, public-private partnerships (PPP), and corporate funding opportunities:

Government Schemes for Smart Solar-Powered Bus Stops in India

Smart Cities Mission (SCM)

- ₹48,000 Cr for urban infrastructure.
- Funds WiFi, digital displays, CCTV, smart ticketing, and solar panels.
- Example: Pune – Solar-powered bus stops with WiFi & digital tracking.

AMRUT (Atal Mission for Rejuvenation and Urban Transformation)

- ₹77,640 Cr for urban mobility improvement.
- Supports public transport, smart bus stops, and renewable energy.
- Example: Indore – Upgraded bus stops with solar lighting & digital boards.

FAME India (Faster Adoption and Manufacturing of Electric Vehicles)

- ₹10,000 Cr for EV infrastructure & smart transport.
- Funds EV charging stations at bus stops.
- Example: Bangalore – BMTC installed EV charging at bus stops.

Renewable Energy Subsidies (MNRE - Ministry of New & Renewable Energy)

- 30%-70% subsidy for solar panel projects.
- Grants available for solar-powered bus stops.
- Example: Chennai – Solar bus stops funded under MNRE.

Gati Shakti National Master Plan

- ₹100 Lakh Cr for transport & smart mobility projects.
- Supports PPP model funding for bus stop modernization.
- Example: Mumbai – Plans to upgrade bus stops under Gati Shakti.

Nirbhaya Fund

- ₹1,000 Cr for women's safety infrastructure.
- Funds CCTV cameras, SOS buttons, and well-lit bus stops.
- Example: Delhi – Installed CCTV & SOS buttons at bus stops.

State-Level Urban Transport Funds

- State-sponsored funds for bus stop modernization & digital ticketing.
- Available in Maharashtra, Karnataka, Tamil Nadu, Delhi, etc.
- Example: Bangalore – Smart bus stops funded by Karnataka transport funds.

PPP Model & CSR Funding

- Govt provides land & approvals, private companies invest in infrastructure.
- CSR funding from Tata, Infosys, Jio, Airtel, Reliance.
- Example: Gurugram – Solar-powered bus stops built through PPP.

Government Permission & Collaboration for Implementation in India

::Permissions Required:

- Municipal Corporation Approval – For land use and construction of bus stops. State Transport Department Approval – For integration with public transport networks.
- Electricity & Renewable Energy Board – For solar panel installation and grid connectivity.
- Urban Development Authority – If part of a Smart City project.
- ::Collaboration Opportunities:
- Public-Private Partnership (PPP) Model – Government provides land & support; private players handle funding & execution.
- Smart City Mission – Collaboration under government initiatives for urban infrastructure.
- CSR Funding – Partner with companies for funding through their Corporate Social Responsibility (CSR) programs

3. Product Description

3.1 Physical Infrastructure

- 10 seats with covered shelter
- Structure: Galvanized steel with polycarbonate/fiber roof

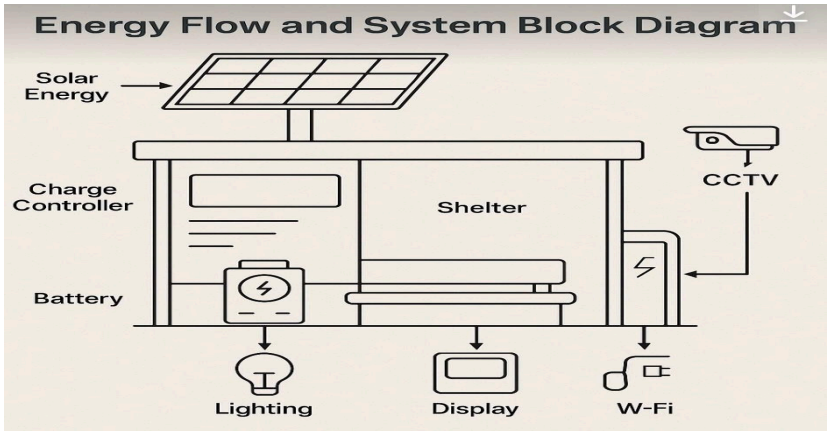
- Accessibility: Ramps, tactile paving

3.2 Smart Components

Component	Function
MPPT Controller	Optimizes battery charging
Solar panels	Electricity generation
Battery bank	Stores energy for night/backup
Wi-Fi Router	Internet access
CCTV camera	24/7 security monitoring
LED lights	Night lighting (auto-on)
USB portals	Mobile charging
Panic Button	Emergency alerts
LED display	Bus schedules, alerts, ads
EV charging port	Charging for Electric Vehicles

4. System Architecture

- Off-grid solar setup with battery bank
- MPPT controller for efficient charging
- Day: solar powers system and charges battery
- Night: system runs on battery
- Automation for lights, internet, surveillance



Schematic diagram of Smart bus stop

Night-time Operation Solution :

- Solar Battery Storage – Use Li-ion or lead-acid batteries to store excess daytime solar energy. Example: Tesla Powerwall or Exide Solar Batteries.
- Motion-Sensor LED Lighting – Automatically turns on when someone is present. Example: Philips Smart LED Motion Sensors.
- Thermal Infrared CCTV Cameras – Provides clear surveillance at night. Example: Hikvision DS- 2TD2617B Thermal Camera.
- Glow-in-the-Dark Bus Route Maps – Solar-powered illuminated signboards. Example: 3M Retroreflective Glow Panels.
- AI-Powered SOS Alert System – Connects emergency buttons to police stations. Example: Securens SOS Alert System.

5. Technical Specifications

5.1 Daily Energy Requirement

Component	Power (W)	Hours	Energy (Wh)
LED Lights (4)	15 × 4 = 60	12	720
Wi-Fi Router	18	24	432
CCTV (2)	15 × 2 = 30	24	720
Digital Display	50	10	500
USB Ports (4)	5 × 4 = 20	7	140
EV Charging Port	2500	7.5	18,750
Total			**21,262 Wh/day ≈ 21.3 kWh/day**

5.2 Solar System Sizing

- Required: 21.3 kWh/day (with buffer)
- Solar Insolation: 4.5 hrs/day
- Panel Needed: 21.3 / 4.5 = ~4.73 kW
- Installed: 10 × 500W panels = 5 kW total
- Battery: 2 days × 21.3 kWh = ~42.6 kWh → 48V, ~900Ah lithium-ion battery bank-
  Required: 2.5 kWh/day (with buffer)
- Solar Insolation: 4.5 hrs/day
- Panel Needed: 2.5 / 4.5 = 555 W
- Installed: 2 x 320W = 640W
- Battery: 2 days x 2.5 kWh = 5 kWh → 12V, ~400Ah

6. Cost Analysis

Item	Qty	Unit Cost (INR)	Total (INR)
Solar Panels (2.5–3 kW total)	8–10 panels	₹20,000	₹1,80,000
Battery Bank (15–20 kWh, Li-ion)	~1200–1600Ah	₹10,000/kWh	₹2,00,000
MPPT Charge Controller (60–80A)	1	₹20,000	₹20,000
Wi-Fi System (with SIM router)	1	₹15,000	₹15,000
CCTV System (High-res DVR + 2 cams)	1	₹20,000	₹20,000
LED Lights (4 × 15W)	4	₹750	₹3,000
USB Charging Ports (Heavy-duty)	1 set	₹7,000	₹7,000
EV Charging Unit (5 vehicles/day)	1	₹80,000	₹80,000
Display Screen (Outdoor grade)	1	₹20,000	₹20,000
Panic Button & Alert System	1	₹5,000	₹5,000
Structure & Seating (10 seats)	1	₹1,20,000	₹1,20,000
Miscellaneous (Install, wiring, permits, civil work)	—	₹50,000	₹50,000
Total Estimated Cost			**₹8,20,000**

7. Payback Analysis

Monthly Revenue/Savings

- Grid savings: ₹800
- Ad revenue: ₹1200
- Total: ₹2000/month

Payback Period

- ₹1.87 lakh / ₹2000 ≈ 7.8 years
- With 40% subsidy → ₹1.1 lakh / ₹2000 = 4.6 years

8. Environmental & Social Impact

- CO_2 Saved: ~700 kg/year
- Supports SDGs 7, 9, 11, 13
- Improves safety and digital access

9. Risks & Mitigation

Risk	Solution
Low sunlight	Extra panel, higher efficiency battery
Vandalism	Tamper-proof enclosures
Wi-Fi Outage	Dual-SIM auto-switch routers

10. Performance Metrics

Metric	Value
Daily Energy Demand	21.3 kWh
Solar Capacity Installed	5 kW
Battery Capacity	42.6 kWh (48V, ~900Ah)
Wi-Fi Coverage Radius	50–70 meters
Seating Capacity	10 persons
Backup Duration (No Sunlight)	2 days
Payback Period (No Subsidy)	7.8 years
Payback Period (With Subsidy)	4.6 years
CO_2 Reduction	~700 kg/year

11. Conclusion

The solar-powered smart bus stop provides a scalable and eco-friendly solution to urban infrastructure needs. It aligns with national goals for sustainable urban development, offering safety, digital inclusion, and energy efficiency with a practical payback period.

12. Reference

Certainly! Here are some authoritative references to support the scalability and revenue models for solar-powered bus shelters:

1. Seedia Smart Solar Bus Shelter

Seedia offers modular, energy-independent bus shelters equipped with solar panels, USB charging, Wi-Fi, and digital displays, showcasing scalable smart city infrastructure.

🔗 https://seedia.city/product/solar-bus-shelter/

2. Yeroo Solar Bus Shelter Features

Yeroo highlights solar-powered shelters with integrated lighting, advertising panels, and charging stations, emphasizing self-sufficiency and environmental sustainability.

🔗 https://www.yeroo.com/article/589/5.html

3. Public-Private Partnership Models for Sustainable Urban Transport

This World Bank report discusses PPP models for urban transport systems, including the development and maintenance of bus shelters through collaborative frameworks.

🔗 https://ppp.worldbank.org/public-private-partnership/sites/default/files/2022-04/Final-Report-PPP-Models-for-Sustainable-Urban-Transport-Systems.pdf

4. Bus Shelter Powered by Solar Market Trends

An analysis of the global market for solar-powered bus shelters, projecting significant growth and highlighting key players and market drivers.

🔗 https://www.linkedin.com/pulse/bus-shelter-powered-solar-market-key-trends-growth-h0kpc/

5. Business Models for Solar Powered Charging Stations

This research paper explores various business models for solar-powered charging stations, relevant for integrating EV charging into bus shelters.

🔗 https://www.researchgate.net/publication/277673682_Business_Models_for_Solar_Powered_Charging_Stations_to_Develop_Infrastructure_for_Electric_Vehicles

These resources provide insights into the design, implementation, and monetization strategies for solar-powered bus shelters, supporting their scalability and integration into smart city initiatives.